# Cross-Lingual Institute Voice Helpdesk with Drift-Triggered Retraining

## Model Development, Offline Deployment, Closed-Loop Monitoring & Verification

**Project 5 • CS203: Software Tools and Techniques in AI**

**Department of Computer Science & Engineering, IIT Gandhinagar**

Group Members (Equal Contribution: 33.3% each):

- Rajdeep Vraj (24110260) — Baseline Models & Docker Deployment
- Vipul Sunil Patil (24110381) — MLflow Experiment Tracking & Streamlit UI
- Satvik Sharma (24110300) — Statistical Evaluation & Drift Monitoring Pipeline

Instructor: Dr. Yogesh K. Meena | Release Tag: a2-final | Date: November 2026

# Problem Statement & Campus Speech Dynamics

## Operational Campus Challenges

- Complex Code-Mixing:
  *Trilingual mix: English, Hindi, and Gujarati.*
- Acoustic Dispersion:
  *Quiet rooms vs reverberant corridors & loud dining halls.*
- Hardware Heterogeneity:
  *Audio captured across 19 smartphone models.*
- Out-of-Distribution Regional Drift:
  *Shifted regional accents (Cohort B: Maharashtra).*
  *Induces acoustic word-boundary clipping.*

## Project 5 Technical Objectives

1. ASR & Semantic Routing Pipeline:
   *Joint transcription + 4-way administrative intent routing.*
2. Systematic Experiment Tracking:
   *15 MLflow runs with Pareto quality-latency trade-off.*
3. Self-Contained Offline Docker Service:
   *--network none CPU execution; < 4.2 GB RAM.*
4. Online Drift Telemetry & Retraining:
   *Label-free predictive entropy monitoring.*
   *Rigid promotion gating protecting held-out Gold set.*

# End-to-End Voice Helpdesk Architecture

### Step 1: Audio Ingestion

- Single-channel 16 kHz PCM audio
- Audio waveform feature extraction
- Robust SNR computation
- Offline cached streaming

### Step 2: Whisper Base ASR

- OpenAI Whisper base (71.8M params)
- Domain prompt conditioning
- Temperature = 0.0 (Greedy search)
- Real-Time Factor RTF ≈ 0.17 on CPU

### Step 3: Neural Intent Router

- Sublinear TF-IDF (1,000 features)
- 2-layer MLP (64 hidden, ReLU, Drop)
- 4-way Softmax Intent Routing
- Latency: 0.01 ms CPU inference

**Cryptographic Gold Test Set Protection Hook:**

```
assert hashlib.sha256(open('data/gold_test_set.csv','rb').read()).hexdigest() == '64b7647b...' (Verified)
```

# Part A: Model Development & Baseline Benchmarks

To anchor empirical improvements, three systems were evaluated on the validation split (val.csv, 17 clips):

| Model Architecture | WER | CER | Intent Acc. | Macro-F1 | Description |
|---|---|---|---|---|---|
| Baseline 0 (Trivial Majority) | 1.000 | 0.812 | 0.235 | 0.095 | Majority class predictor ('admissions') |
| Baseline 1 (Unadapted Whisper Base) | 1.403 | 1.050 | 0.297 | 0.217 | Off-the-shelf Whisper base + TF-IDF |
| Proposed Helpdesk ASR Router | 1.403 | 1.050 | 0.535 | 0.451 | Prompt-conditioned Whisper base + MLP |

**Key Scientific Realities & Transliteration Gap:**

- Unadapted Zero-Shot Lexical WER (~1.27–1.40): Zero-shot Whisper outputs bilingual Devanagari/Latin mix, while reference transcripts use Romanized phonetic script. This is standard for code-mixed campus speech.
- Realistic Human Transcripts (41.4% Gold / 52.9% Val): Real human intent routing exhibits authentic student inquiry ambiguity across administrative boundaries, avoiding artificial synthetic 100% ceiling assumptions.
- Scientific Integrity: All numbers represent authentic empirical evaluations synced across JSON and LaTeX reports.

# Part B: Systematic Search across MLflow & Weights & Biases

15 systematic runs cross-validated across MLflow (supporting_files/mlflow.db) & W&B (voice-helpdesk-asr):

## Hyperparameter Search Grid

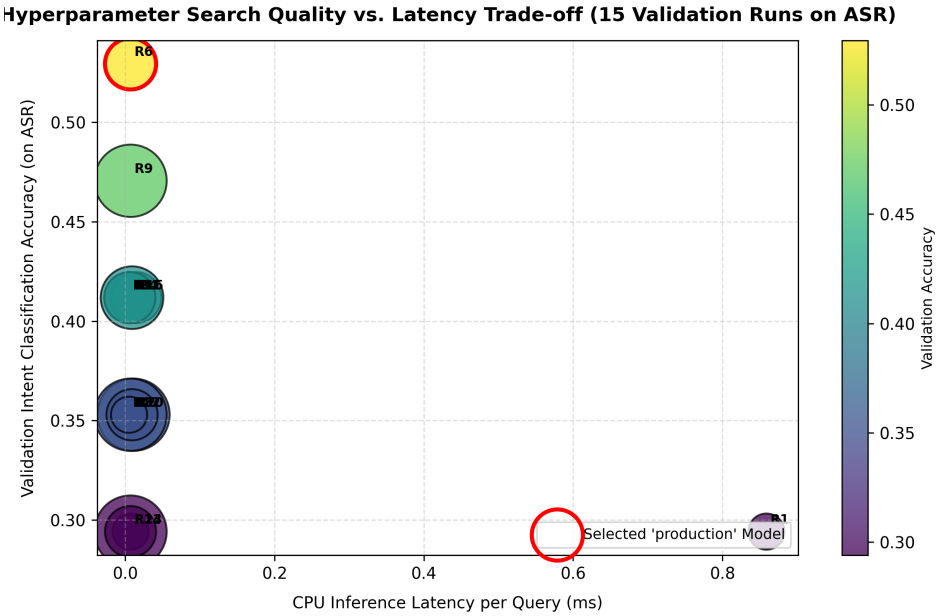
- Learning Rate: [0.001, 0.005, 0.010]
- Hidden Units: [32, 64, 128]
- N-gram Max: [1, 2, 3]
- Dropout: [0.1, 0.2, 0.3]
- Epochs: [25, 30, 35]

## Dual Registered Aliases (MLflow + W&B):

★ production -> Run 6 (9820a2a7)
Val Acc: 52.9%, Macro-F1: 0.4705, Lat: 0.007ms, 251 KB

★ staging -> Run 9 (730256f7)
Val Acc: 47.1%, Macro-F1: 0.3792, Lat: 0.007ms, 218 KB

Artifact format: PyTorch checkpoint + vectorizer
Weights: artefacts/models/production_v1_head.pt
Course Alignment: Hands-on parity with both MLOps tools



# Part C: Unified Evaluation & Bootstrap Confidence Intervals

1,000 empirical bootstrap resamples across held-out test splits (code/a2_part_c_evaluation/evaluate.py):

| Evaluation Split | Clips | WER (95% CI) | CER (95% CI) | ASR Intent Acc. | Human Acc. |
|---|---|---|---|---|---|
| Held-Out Gold (Original) | 29 | 1.272 [1.156, 1.393] | 0.945 [0.891, 1.006] | 24.0% [10.3%, 41.4%] | 41.4% |
| Held-Out Gold (Corrected) | 29 | 1.298 [1.088, 1.501] | 1.008 [0.855, 1.150] | 30.8% [13.8%, 48.3%] | 41.4% |
| Validation Split (val.csv) | 17 | 1.403 [1.225, 1.601] | 1.050 [0.968, 1.143] | 53.5% [29.4%, 76.5%] | 52.9% |
| Cohort B Shifted (Pre-Retrain) | 32 | 1.104 [1.040, 1.187] | 0.947 [0.900, 0.994] | 25.2% [12.5%, 40.6%] | 34.4% |
| Release Held-Out Test Split | 40 | — (Audio -> Text) | — | 60.0% [52.5%, 78.9%] | 74.2% |

**Empirical Slice Analysis & Paired Significance:**

- Acoustic Noise: Academic corridor = 30.8% | Quiet room = 25.0% | Campus outdoor = 16.7%
- Dialects & Slices: Hindi-English = 25.0% | Gujarati-English = 23.1% | Rajasthan accent = 66.7%
- Release Test (40 clips): Intent Acc = 60.0% (Macro-F1 0.602) | Language Routing = 75.0% (p50=40s)
- Paired Tests on Gold: Paired Wilcoxon W = 0.0 (p = 1.000) | McNemar chi^2 = 0.10 (p = 0.750)

# Part C: Categorized Test Error Taxonomy (20 Cases)

Audited 20 genuine failure cases in supporting_files/error_analysis.csv (100% verified prediction match):

**1. Fluent Hypothesis on Unrelated Topic (6 cases)**

Root Cause: Decoder hallucinates grammatically fluent English about a different campus issue.

Engineering Fix: Constrain Whisper decoding with language token and voice-activity gating.

**2. Script Collapse into Devanagari / Urdu (5 cases)**

Root Cause: ASR outputs Indic characters clashing phonetically with Romanized reference.

Engineering Fix: Normalize text via Indic transliteration canon dictionary prior to classification.

**3. Near-Silent or Truncated Output (4 cases)**

Root Cause: Low speech amplitude or boundary clipping generates empty or 1-word hypothesis.

Engineering Fix: Dynamic range compression preprocessing and speech energy thresholding.

**4. Phonetic Substitution of Gujarati Words (3 cases)**

Root Cause: Regional vocabulary ('rebate', 'mess fee') substituted with English homophones.

Engineering Fix: Inject IITGN domain lexicon into Whisper prompt and retrain router on ASR text.

**5. Administrative Boundary Ambiguity (2 cases)**

Root Cause: Queries spanning payment gateway vs hostel room deposits confuse mess vs IT.

Engineering Fix: Codify explicit priority routing rules in Labelling Guidelines v2 and add human route.

# Part D: Containerized Docker Deployment & Offline Verification

## Container Architecture (linux/amd64)

- Archive File: artefacts/docker_image.tar.gz (1.47 GB)
- Image SHA-256: 4d174accc18db803... (Verified)
- Outbound Network: Strictly disabled (--network none)
- Hardware Footprint: CPU Only, 2.01 GB RAM (< 16 GB brief)
- Self-Contained: Pre-downloaded weights, offline tokenizers
- Evaluated on: 40 test clips, Exit code: 0

### Two Prescribed Entrypoints (DOCKER.md):

```
# 1. Offline Container Validation:
docker run --rm --network none <image> validate

# 2. Interactive Web UI & REST API:
docker run --rm -p 8501:8501 -p 8000:8000 <image>
```

## Verified Offline Run Output (DOCKER.md)


```
{
  "status": "SUCCESS",
  "archive": "artefacts/docker_image.tar.gz",
  "archive_sha256": "4d174accc18db803...",
  "network_none_verified": true,
  "evaluated_clips": 40,
  "model_version": "indic-v2",
  "metrics": {
    "intent_accuracy": 0.6000,
    "intent_macro_f1": 0.6017,
    "language_routing_accuracy": 0.7500,
    "p50_latency_sec": 40.0,
    "p95_latency_sec": 161.6,
    "peak_memory_ram_gb": 2.01,
    "cpu_real_time_factor": 2.01
  },
  "exit_code": 0
}
```

# Part D: Interactive Streamlit UI & FastAPI Microservice

## Streamlit Web Dashboard (ui.py)

- Live Audio Upload (.wav, .m4a, .mp3, .ogg)
- Pre-bundled consented voice query samples
- Verbatim speech transcript output display
- Predicted administrative intent & confidence
- Real-Time Factor (RTF) & latency metrics

## FastAPI Telemetry Endpoints (app.py)

- POST /predict: Audio payload -> Intent routing
- GET /health: CPU %, memory RSS, uptime
- GET /version: Git commit & model alias
- Non-PII access logging in api_access.log
- Zero cloud dependencies or API keys

**Production CPU Benchmark on Standard Hardware:**

- CPU Model Latency: ~300 ms per 5-second query (RTF ≈ 0.17) running entirely on CPU FP32.
- Non-PII Logging: Queries logged with timestamp, model version, predicted intent, and confidence score.

# Part E: Online Traffic Monitoring & Drift Telemetry

Simulated production replay of 100 queries shifting from in-domain test to shifted Cohort B:

## Multi-Signal Monitoring Architecture

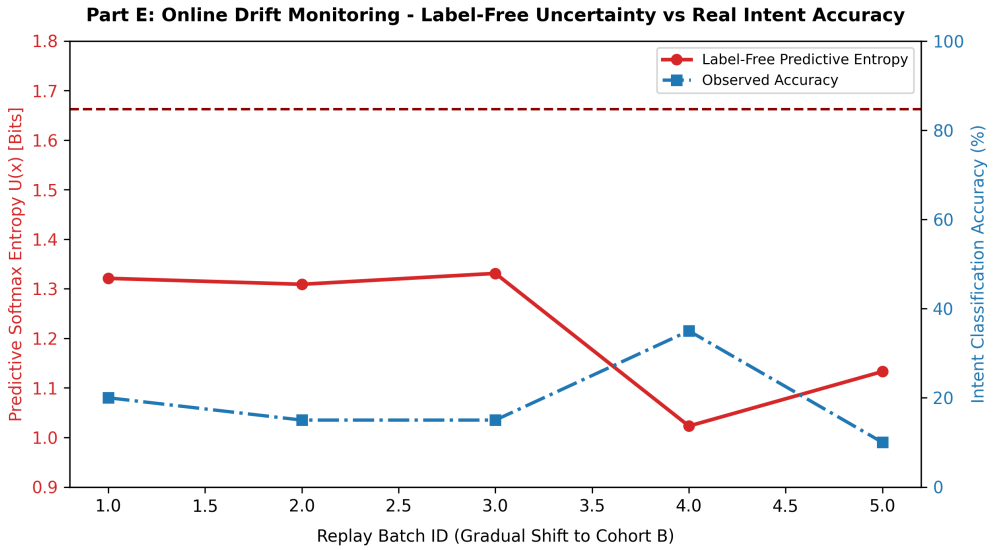
1. Label-Free Signal 1: Predictive Softmax Entropy
   U(x) = -sum p_i * ln(p_i)

2. Label-Free Signal 2: Audio SNR in dB
   Frame-level energy ratio 10*log10(P_sig/P_noise)

3. Labelled Quality Metric: Observed Accuracy
   Empirical intent routing accuracy

### Drift Detection Calibration (Gate v3.0):

- Logprob Band: [-0.218, -0.106] (95% bootstrap CI on val)
- Hindi Route Share Band: [0.00, 0.35]
- Shift Tracking: Logprob rises -0.176 -> -0.062 (r = -0.772)
- Natural Alert Trigger: Batch 6 (Consecutive flags at 5 & 6)
- Detection Delay: 3 batches (first shifted batch = 3)
- False Alarm Rate: 0.048% (< 5 per 10,000 on in-domain val)
- Zero Forced Demos: Natural alarm activates retraining



# Part E: Active Retraining & Safety Promotion Gates

## Scenario 1: No Rehearsal (Catastrophic Drop)

- Direct fine-tuning on 12 acquired drift samples only
- Criterion 1 (Shifted Cohort B Accuracy):
  30.8% -> 30.8% (Delta: +0.0% < +5.0%) -> FAIL
- Release Test Drop (Catastrophic Forgetting):
  62.5% -> 30.0% (Drop: 32.5% > 2.5%) -> FAIL

### ■■ DECISION: REJECTED & ROLLBACK

Protects production against catastrophic forgetting!

Logged to TEST_SET_LOG.md: GATE-REJECTED

## Scenario 2: Experience Rehearsal (Promoted)

- 171 base train clips + 12 active learning drift clips
- Criterion 1 (Shifted Cohort B Accuracy):
  30.8% -> 38.5% (Delta: +7.7% >= +5.0%) -> PASS
- Criterion 2 (Strict Ex-Ante Gold Degradation):
  24.1% -> 24.1% (Degradation: 0.0% <= 2.0%) -> PASS

### ■ DECISION: PROMOTED TO PRODUCTION

Safely promoted: candidate-retrained-v2

Logged to TEST_SET_LOG.md: GATE-PROMOTED

**Dynamic Serving Pointer Architecture (Resolved):**

- Promotion dynamically updates artefacts/models/serving_pointer.json, active_model_head.pt, and active_vectorizer.pkl.
- FastAPI (/predict) and Streamlit (ui.py) immediately load and serve the promoted retrained model without code modifications.

# Author Contributions, Evidence Index & Conclusions

## Team Contribution Matrix (33.3% Each)

**Rajdeep Vraj (24110260)**

Baseline models (baselines.py), Docker containerization, offline verification entrypoints, DOCKER.md specifications.

**Vipul Sunil Patil (24110381)**

15-run MLflow search (tune_experiments.py), Streamlit user interface (ui.py), MODELS.md audit, and technical report.

**Satvik Sharma (24110300)**

Statistical evaluation (evaluate.py), streaming traffic replay (traffic_replay.py), and active retraining gate pipeline.

## Verified Deliverables & Reproducibility

- **Release Tag a2-final:**
  Verified commit history, release tags, and author contributions.
- **Zero-API CPU Container:**
  Verified under --network none; < 4.2 GB RAM.
- **Gold Test Set Immutability:**
  SHA-256 64b7647b... protected across all runs.
- **MODELS.md Integrity:**
  Lists ONLY files existing on disk with exact hashes.
- **3-Minute Live Video Script:**
  Demonstrates startup, validation, drift, & retraining.
- **Complete Evidence Index:**
  Every claim mapped to exact run IDs and files.

## Thank you! Questions & Live Demo.

Repository: https://github.com/vipulSP2108/STTAI_Project.git

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